

KRANTHI KUMAR REDDY ANNAPUREDDY – SOFTWARE DEVELOPER

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PROJECTS

Core Technologies: Java, Spring Boot, Spring Security, Git, Docker, Maven, Gradle, MySQL, PostgreSQL

Core Tools: IntelliJ, Postman, VSCode, DBeaver

Gateway

October 2025 - Present

Technologies: Reactive WebFlux

- Designed a single entry point using **Spring Cloud Gateway** to handle dynamic routing, path predicates, and load balancing from an **Angular** frontend to downstream microservices.
- Configured the gateway as an **OAuth2 Resource Server** to validate **JWT** tokens at the edge, mitigating unauthorized access and abstracting the internal microservice topology.
- Implemented cookie-based **CSRF protection** tailored for reactive architectures, seamlessly syncing CSRF tokens with the client to prevent cross-site request forgery attacks.
- Eliminated configuration bloat by centralizing cross-origin resource sharing (**CORS**) and error mitigation policies directly at the gateway layer.

Authorization Server

January 2026 - Present

Technologies: JJWT, Cryptography

- Built a custom authentication server using `io.jsonwebtoken`, leveraging a **PKCS12** KeyStore to handle secure **RSA** public/private key cryptographic signing.
- Exposed a public **JWKS** endpoint (`/.well-known/jwks.json`) to allow the API Gateway and downstream resource servers to independently fetch public keys and verify **JWT** validity.
- Engineered a **multi-tenant** authentication framework that injects tenant context directly into JWT claims, enforcing data isolation at the perimeter.
- Developed an enterprise **RBAC** engine mapping custom roles and explicit privileges to users, establishing strict claim-based authorization across the ecosystem.

Data Management

December 2025 - Present

- Engineered an Angular dashboard integrated with **HTTP Interceptors** to automatically inject OAuth2 Bearer Tokens into API requests and manage global error states.
- Built an **OAuth2 Resource Server** utilizing **JWKS** for asymmetric JWT token verification, enforcing **tenant-based** isolation and Role-Based Access Control (**RBAC**) across a microservices ecosystem.
- Eliminated manual production **SQL** updates by building a validated API layer with automated user auditing.
- Protected database connection pools by disabling **OSIV**, cleanly resolving lazy-loading risks by mapping **JPA** entities to Java Records within **@Transactional** service boundaries.
- Extracted core data models and custom exceptions into a version-controlled, shared enterprise utility library published on **GitHub**.
- Leveraged **Angular Signals** to manage fine-grained, reactive UI state across the data dashboard, optimizing rendering performance and eliminating complex RxJS subscription overhead for synchronous data flows.

Portfolio

June 2025 - Present

- Automated **CI/CD** using **GitHub Actions**, reducing develop-to-deploy time by 80%.
- Deployed a **Dockerized API** on **Render**, powering a **GitHub Pages**-hosted Angular application with live data.
- Configured backend as an **OAuth2 Resource Server** to handle secure token validation and endpoint authorization via an external auth server.
- Designed a secure, decoupled full-stack infrastructure integrating a managed **PostgreSQL** database with **Spring Data JPA**.
- Engineered a high-resilience frontend data layer using **Angular Signals**, implementing a custom **HTTP interceptor** with dynamic timeout scaling (up to 90s) to gracefully handle backend cloud container latency while maintaining an uninterrupted **UX**.
- Optimized application performance and network overhead by designing a single-query unified payload strategy, caching crucial system state reactively to minimize redundant downstream **API** pressure.
- Leveraging **Lombok** to reduce boilerplate and implementing structural logging configurations for long-term production auditing.

Shared Library

March 2026 - Present

- Designed, developed, and published a decoupled, version-controlled library to **GitHub Packages**, enforcing strict DRY principles and eliminating redundant boilerplate across enterprise microservices.
- Designed a zero-configuration auto-configure architecture. Dynamically separated bean definitions via conditional annotations for both **SERVLET**, **REACTIVE** environments, enabling seamless single-dependency activation.
- Built a unified, cross-stack error-handling hierarchy and a custom abstract `BaseException` class (extends **RuntimeException**), standardizing global, immutable error payloads across all hosting applications.
- Standardized data transport layers by leveraging **Java 17 Records** alongside reusable utility classes and custom validation groups (**CreateValidation**, **UpdateValidation**) to enforce consistent runtime request parsing.
- Maintained strict library decoupling using `<optional>true</optional>` scopes.

EDUCATION

Master of Engineering in Internetworking

SEPT 2023 - MAY 2025

Dalhousie University, Halifax, NS

- Relevant Courses: Programming in Internetworking (Python, Docker, CI/CD pipelines).

Bachelor of Engineering in Electrical and Electronics Engineering

JUN 2016 - SEPT 2020

Anna University, R.M.D. Engineering College, India

- Relevant Courses: Object Oriented Programming (C, C++, Java languages and Algorithms).

EXPERIENCE

Freelancer

SEPT 2024 - PRESENT

Software Developer

- Accelerated deployment times by 40% by migrating a legacy **monolithic application** to a highly scalable, **RESTful microservices architecture**.
- Documented **Module Federation** based **Micro Frontend architecture** for an Angular application, providing guidelines for integration, configuration, and module sharing across applications.
- Designing and compiling comprehensive architectural **documentation** for the microservices ecosystem, mapping out end-to-end data flows across the API Gateway, custom OAuth2/JWKS Authorization Server, and multi-tenant resource servers to create production-ready engineering blueprints.
- Collaborated with **UI/UX teams** to translate wireframes and business requirements into intuitive, user-friendly web interfaces.

SoWeDid Software Solutions, Hyderabad, India

OCT 2020 - MAY 2023

Software Developer

- Refactored outdated codebases to adopt modern frameworks and design patterns, enhancing maintainability and reducing technical debt.
- Built and optimized **Elasticsearch queries and indexes**, reducing search response time by 30% and supporting faster data retrieval across applications.
- Designed and optimized **BIRT reports**, improving performance and delivering insights to 100+ business users with reduced report generation time.
- Implemented and customized Angular DataTables to handle large datasets efficiently, delivering advanced features like server-side pagination, multi-column sorting, and dynamic filtering for end-users.
- Integrated **RESTful APIs** with Angular application, using **JSON**, enabling seamless data flow and real-time updates for end-users.
- Maintained session-based authentication using **Spring Security and HTTP Sessions** and login workflows with an **Angular application** for secure user access.
- Documented **Liquibase** workflows and best practices, making it smooth on-boarding for the new team members and improving database release cycle by 25%.
- Worked in **Agile** development environments with regular sprint planning, code reviews and collaborative development workflows.
- Investigated and resolved software issues by analyzing root causes and implementing effective solutions.
- Quickly learned new technologies and adapted to evolving project requirements.